

CBSE Test Paper 02
Chapter 11 Thermal Properties of Matter

1. If the substance is in the form of a long rod, then for small change in temperature, ΔT , the fractional change in length, $\Delta l/l$, is **1**
 - a. directly proportional to $\log \Delta T$
 - b. directly proportional to ΔT
 - c. inversely proportional to ΔT
 - d. inversely proportional to $\log \Delta T$
 2. The coefficient of volume expansion is n times coefficient of linear expansion where n equals **1**
 - a. 4
 - b. 2
 - c. 3.0
 - d. 5
 3. In an experiment on the specific heat of a metal, a 0.20 kg block of the metal at 150°C is dropped in a copper calorimeter (of water equivalent 0.025 kg) containing 150 cm^3 of water at 27°C . The final temperature is 40°C . Compute the specific heat of the metal. **1**
 - a. $0.43\text{ J g}^{-1}\text{K}^{-1}$
 - b. $0.37\text{ J g}^{-1}\text{K}^{-1}$
 - c. $0.40\text{ J g}^{-1}\text{K}^{-1}$
 - d. $0.46\text{ J g}^{-1}\text{K}^{-1}$
 4. A water heater is operated by solar power. If the solar collector has an area of 6.00 m^2 and the power delivered by sunlight is 550 W/m^2 , how long does it take to increase the temperature of 1.00 m^3 of water from 20.0°C to 60.0°C ? **1**
 - a. 50.7 ks
 - b. 54.7 ks
 - c. 56.7 ks
 - d. 52.7 ks
 5. The triple points of neon is 24.57 K. Express this temperatures on the Celsius and
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Fahrenheit scales 1

- a. $-208.58^{\circ}\text{C} = -355.44^{\circ}\text{F}$
 - b. $-238.58^{\circ}\text{C} = -315.44^{\circ}\text{F}$
 - c. $-248.58^{\circ}\text{C} = -415.44^{\circ}\text{F}$
 - d. $-218.58^{\circ}\text{C} = -335.44^{\circ}\text{F}$
6. The earth without its atmosphere would be inhospitably cold. Why? 1
 7. Why an ice box is constructed with a double wall? 1
 8. Metal disc has a hole in it. What happens to the size of the hole when disc is heated? 1
 9. On a hot day, a car is left in sunlight with all the windows closed. After some time, it is found that the inside of the car is considerably warmer than the air outside. Explain, why? 2
 10. 100 g of water is super cooled to -10°C . At this point, due to some disturbance mechanized or otherwise some of it suddenly freezes to ice. What will be the temperature of the resultant mixture and how much mass would freeze?
[$S_w = 1\text{cal/g/}^{\circ}\text{C}$ and $L_{fusion}^W = 80\text{cal/g}$] 2
 11. a. Why the brake drums of a car are heated when it moves down a hill at constant speed?
b. Why pendulum clocks generally go faster in winter and slow in Summer? 2
 12. A thin rod having length L_0 at 0°C and coefficient of linear expansion α has its two ends maintained at temperatures θ_1 and θ_2 , respectively. Find its new length. 3
 13. Two vessels A and B of different materials but having identical shape, size and wall thickness are filled with ice and kept at the same place. Ice melts at the rate of 100 g m in⁻¹ and 150g min⁻¹ in A and B, respectively. Assuming that heat enters the vessels through the walls only, calculate the ratio of thermal conductivities of their materials. 3
 14. A steel tape 1m long is correctly calibrated for a temperature of 27.0°C . The length of a steel rod measured by this tape is found to be 63.0 cm on a hot day when the temperature is 45.0°C . What is the actual length of the steel rod on that day? What is the length of the same steel rod on a day when the temperature is 27.0°C ? Coefficient of linear expansion of steel = $1.20 \times 10^{-5} \text{ K}^{-1}$. 3
 15. A body cools from 80°C to 50°C in 5 minutes. Calculate the time it takes to cool from 60°C to 30°C . The temperature of the surroundings is 20°C . 5
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